



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 02

Experiment Title: Design and Simulation of 8-bit Full Adder using Verilog HDL

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1. Experiment Name

Design and Simulation of 8-bit Full Adder using Verilog HDL

2. Objective

The objectives of this experiment are:

- To design an 8-bit Full Adder circuit using Verilog HDL.
- To understand the operation of combinational arithmetic circuits.
- To generate RTL schematic of the designed circuit using Xilinx ISE.
- To simulate and verify the functionality of the Full Adder circuit.
- To analyze the Sum and Carry output for different input combinations.

3. Apparatus / Hardware & Software Requirements

Software Requirements:

- Xilinx ISE Design Suite 14.7
- ISim Simulator
- Verilog HDL

Hardware Requirements:

- Computer system

4. Circuit Diagram / Schematic

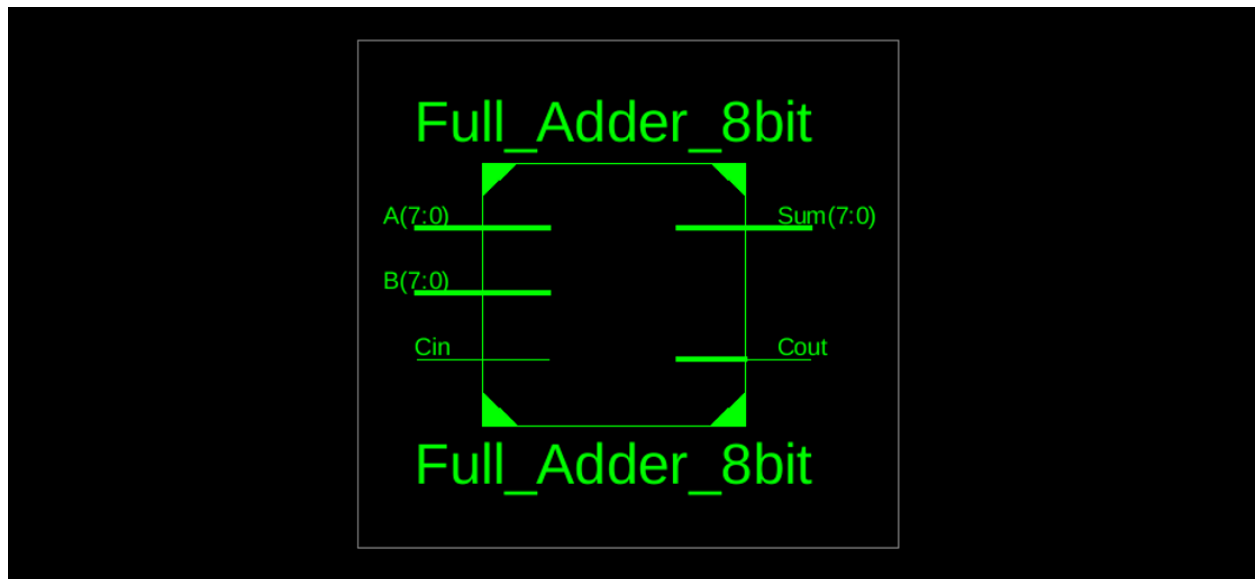


Figure 01: Block-Level RTL Schematic of 8-bit Full Adder

The block-level RTL schematic represents the 8-bit Full Adder as a single functional unit. It contains two 8-bit inputs (A and B), one carry input (Cin), and two outputs (Sum and Cout).

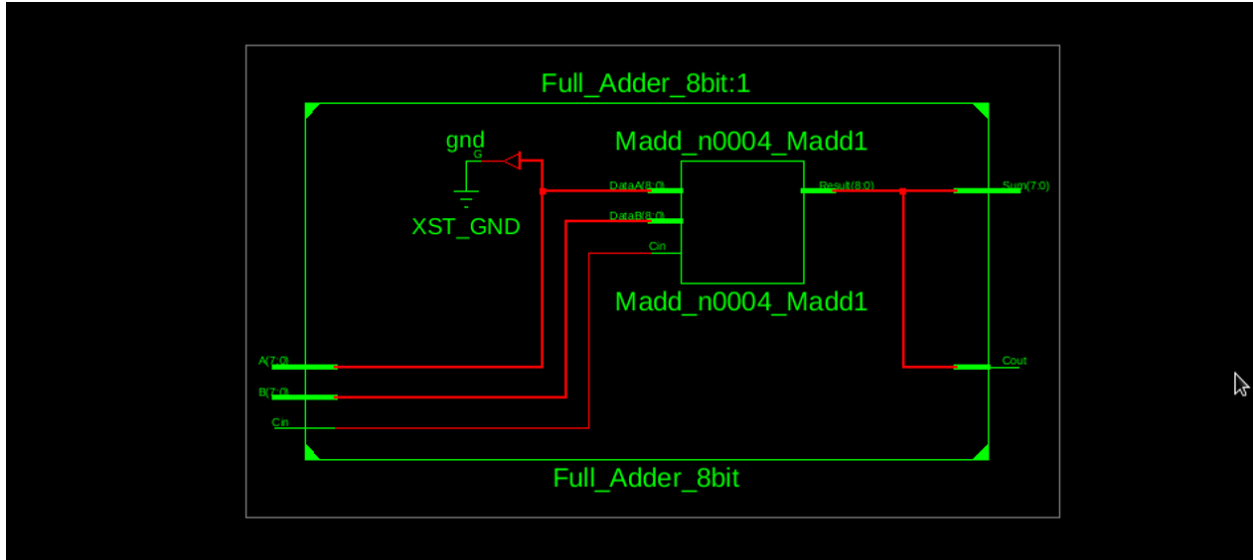


Figure 2: Synthesized RTL Schematic of 8-bit Full Adder

The synthesized RTL schematic shows the optimized internal implementation generated by Xilinx synthesis tools.

6. Output Analysis / Observations

The designed 8-bit Full Adder was simulated using ISim simulator. Different input combinations were applied to verify the output.

Input A	Input B	Cin	Sum	Cout
00	00	0	00	0
FF	01	0	00	1
0F	01	0	10	0
55	AA	0	FF	0
9D	64	0	01	1
10	20	1	31	0

Table 01: Simulation Test Cases and Output Verification of 8-bit Full Adder

The simulation waveform confirmed that the output values matched the expected binary addition results.

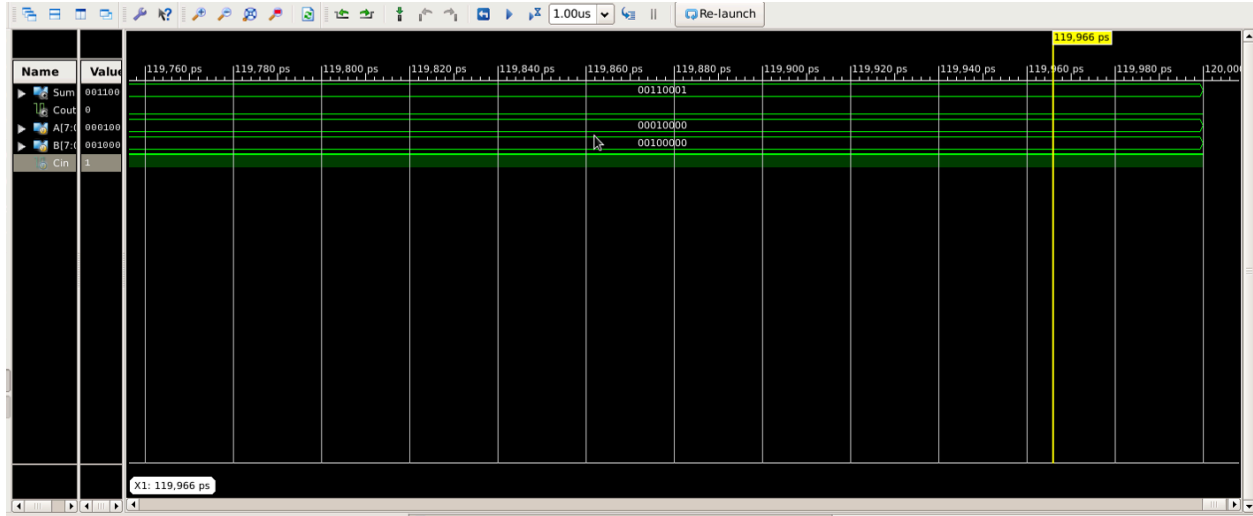


Figure 03: Simulation Waveform 8bit

7. Result

The 8-bit Full Adder circuit was successfully designed and simulated using Verilog HDL in Xilinx ISE. The simulation results verified that the Sum and Carry outputs were generated correctly for different input combinations.

Therefore, the objective of the experiment was successfully achieved.

8. Github Repository Commits

GitHub Repository: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/tree/main/Full_Adder_8bit

Commit Link: https://github.com/Sakhawat025/CSE-450_VLSI_Lab/commits/main/Full_Adder_8bit

9. Conclusion

In this experiment, an 8-bit Full Adder was successfully designed using Verilog HDL and implemented using Xilinx ISE. The RTL schematic and simulation waveform were analyzed to verify the correct operation of the circuit. The tested input combinations produced accurate Sum and Carry outputs, which confirmed the successful completion of the experiment.